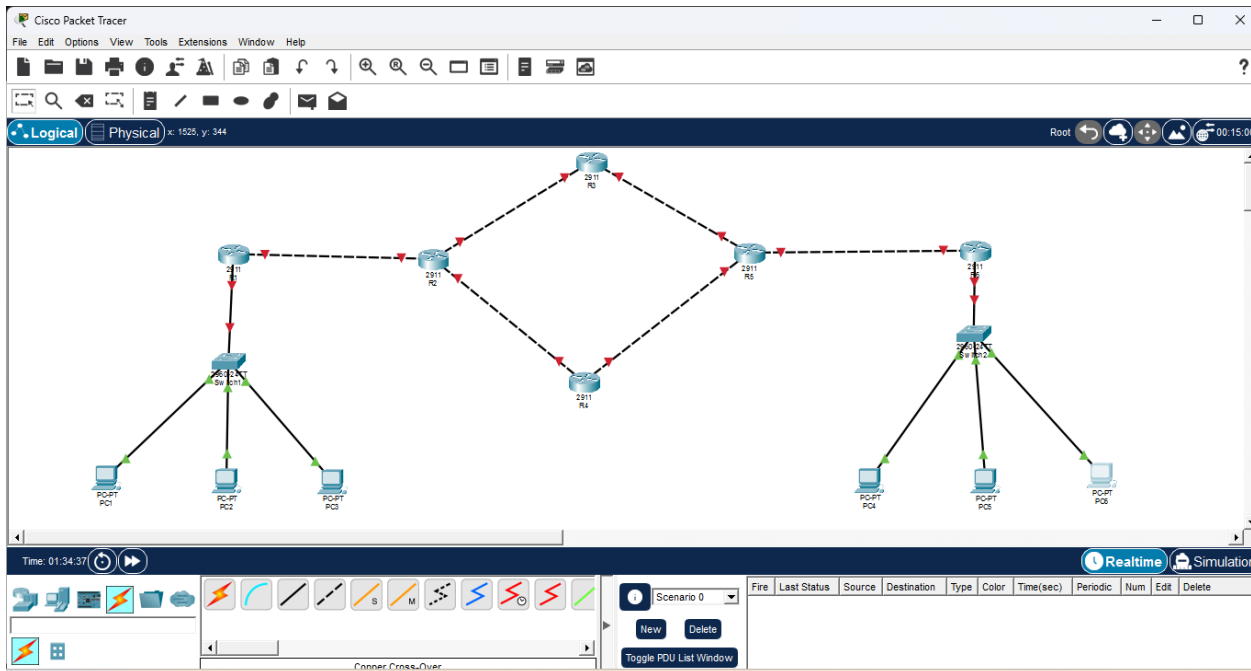


Static Routing

Static routing is a manually configured networking method where an administrator explicitly defines the paths that data packets take to reach specific network destinations.



IP assign for PC1

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::203:E4FF:FE58:7EB9

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

IP assign for PC2

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::210:11FF:FE75:5197

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

IP assign for PC3

PC4

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.20.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:96FF:FE00:5C7D

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Top

IP assign for PC4

PC3

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.4

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:D3FF:FE9E:2871

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Top

IP assign for PC5

PC5

Physical Config **Desktop** Programming Attributes

IP Configuration x

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.20.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::203:E4FF:FE1A:C280

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Top

IP assign for PC6

PC6

Physical Config **Desktop** Programming Attributes

IP Configuration x

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.20.4

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2E0:B0FF:FE98:3C37

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

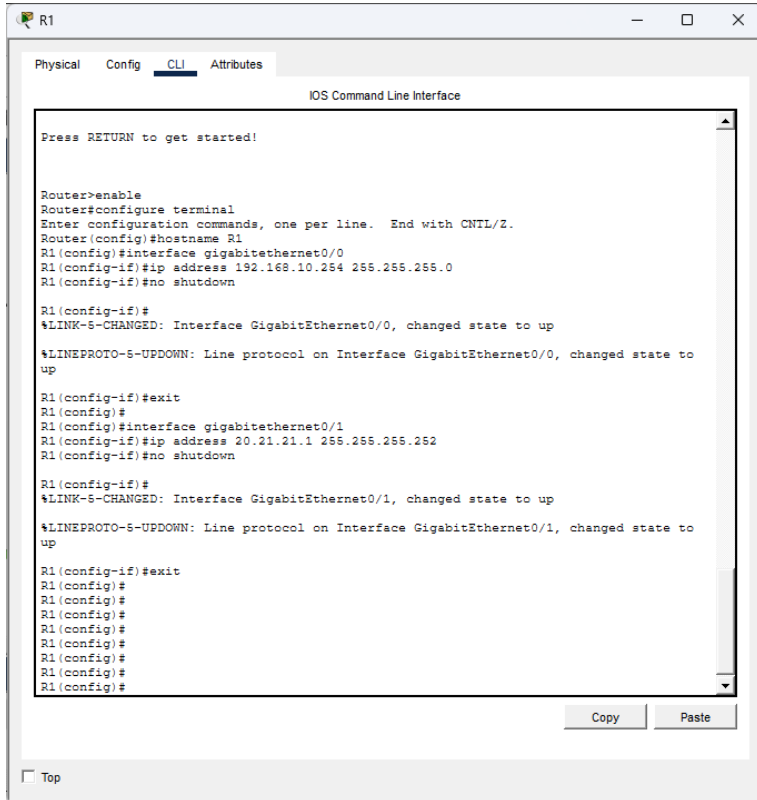
Authentication MD5

Username

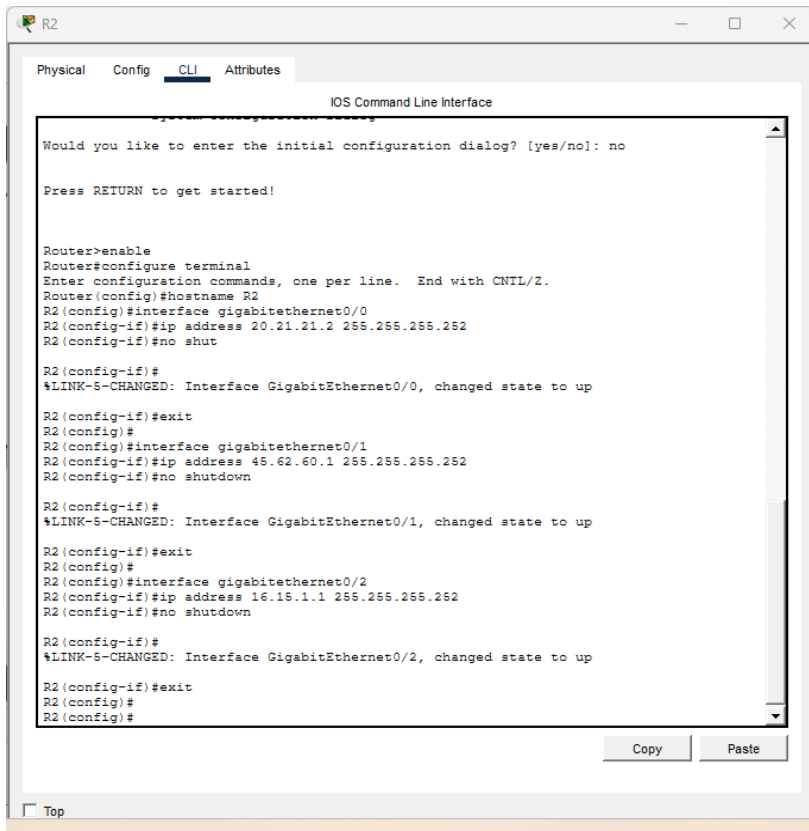
Password

Top

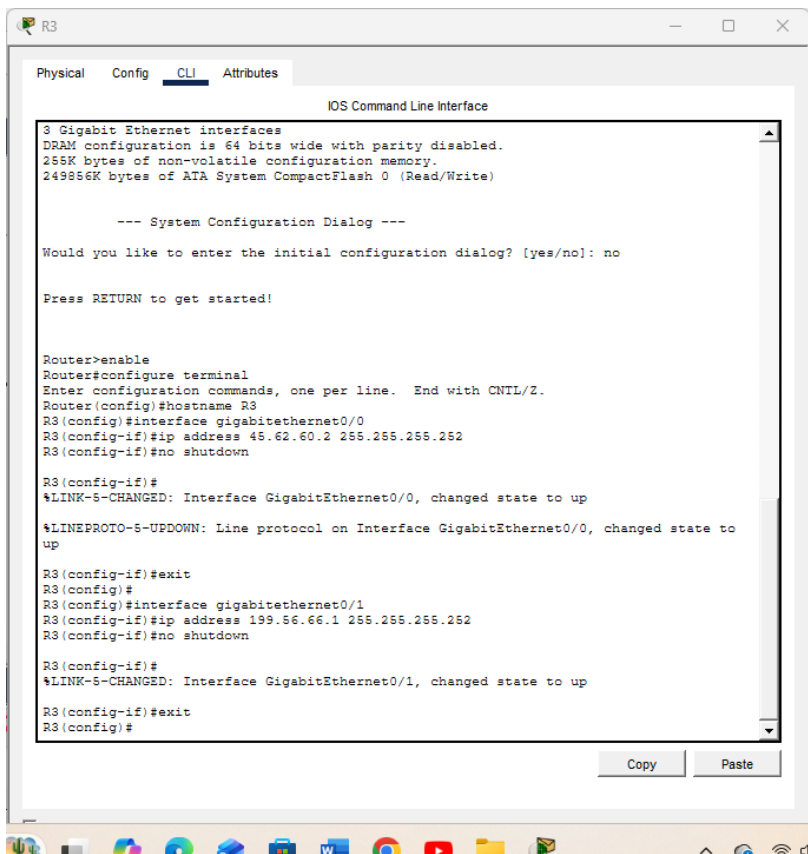
R1 Configure



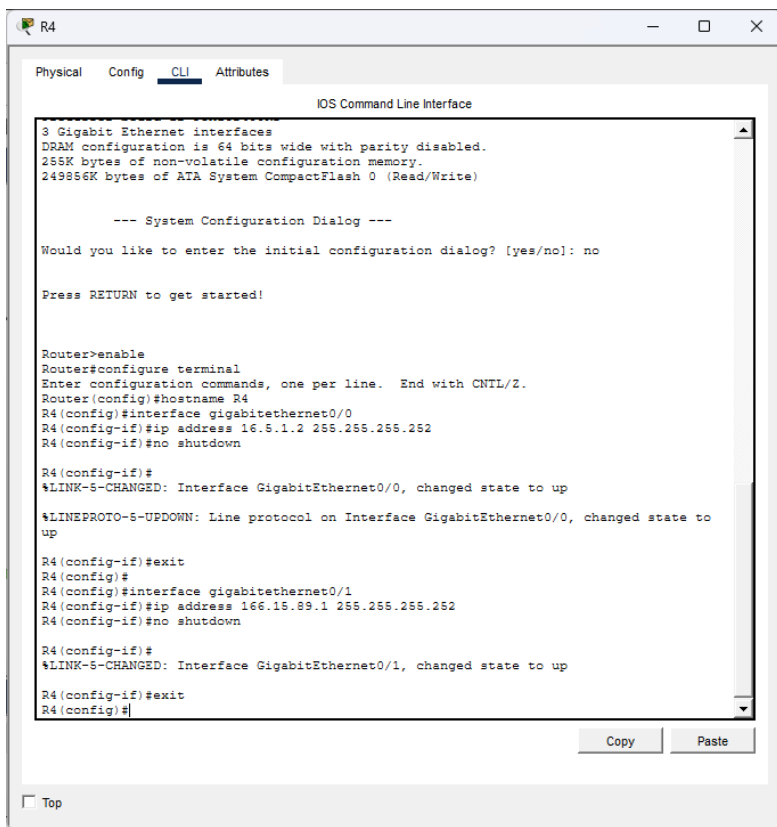
R2 Configure



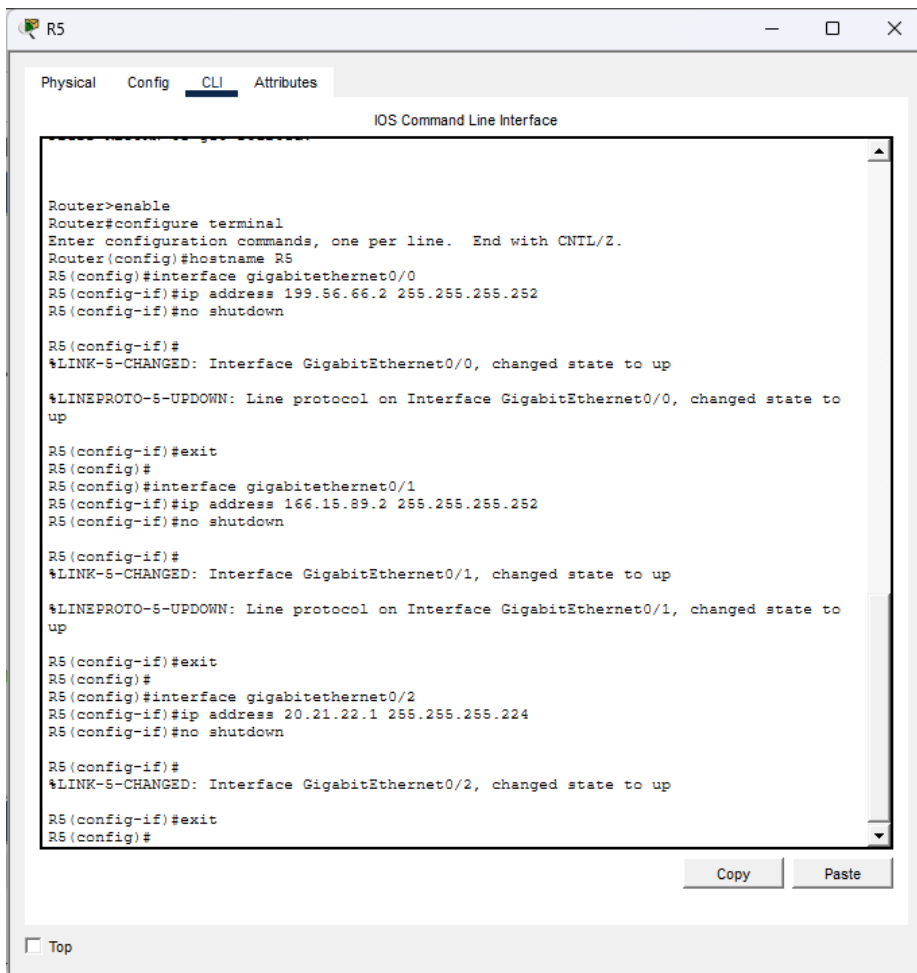
R3 Configure



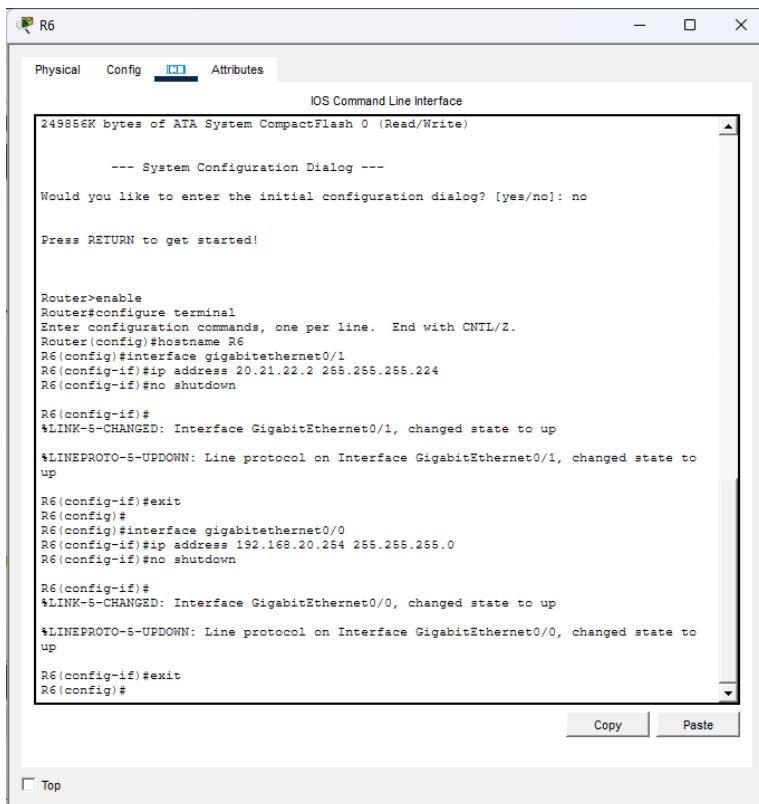
R4 Configure

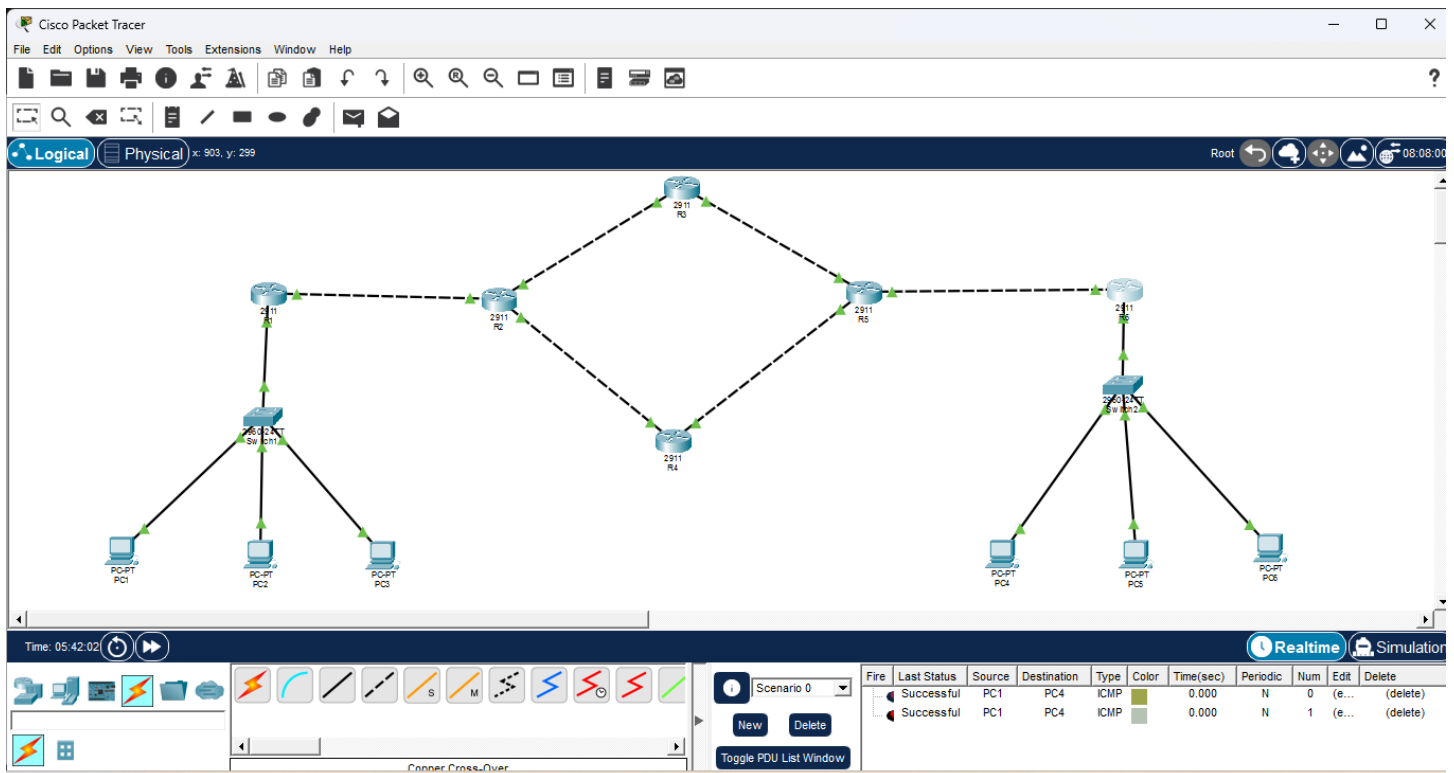


R5 Configure



R6 Configure





Static routing for R1

```

R1
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started.

R1>enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip route 192.168.20.0 255.255.255.0 20.21.21.2
R1(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

  20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.21.21.0/30 is directly connected, GigabitEthernet0/1
L    20.21.21.1/32 is directly connected, GigabitEthernet0/1
  192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0
L    192.168.10.254/32 is directly connected, GigabitEthernet0/0
S    192.168.20.0/24 [1/0] via 20.21.21.2
R1(config)#
  
```

Static routing for R2

R2

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

R2>enable
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip route 192.168.20.0 255.255.255.0 45.62.60.2
R2(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

 16.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    16.15.1.0/30 is directly connected, GigabitEthernet0/2
L    16.15.1.1/32 is directly connected, GigabitEthernet0/2
C    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.21.21.0/30 is directly connected, GigabitEthernet0/0
L    20.21.21.2/32 is directly connected, GigabitEthernet0/0
C    45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    45.62.60.0/30 is directly connected, GigabitEthernet0/1
L    45.62.60.1/32 is directly connected, GigabitEthernet0/1
S    192.168.20.0/24 [1/0] via 45.62.60.2

R2(config)#
```

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☐ Top

R2

Physical Config CLI Attributes

IOS Command Line Interface

```
R2>enable
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip route 192.168.10.0 255.255.255.0 20.21.21.1
R2(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

 16.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    16.15.1.0/30 is directly connected, GigabitEthernet0/2
L    16.15.1.1/32 is directly connected, GigabitEthernet0/2
C    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.21.21.0/30 is directly connected, GigabitEthernet0/0
L    20.21.21.2/32 is directly connected, GigabitEthernet0/0
C    45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    45.62.60.0/30 is directly connected, GigabitEthernet0/1
L    45.62.60.1/32 is directly connected, GigabitEthernet0/1
S    192.168.10.0/24 [1/0] via 20.21.21.1
S    192.168.20.0/24 [1/0] via 45.62.60.2

R2(config)#
```

Copy Paste

☐ Top

Static routing for R3

R3

Physical Config CLI Attributes

IOS Command Line Interface

```
R3(config)#ip route 192.168.20.0 255.255.255.0 199.56.66.2
R3(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    45.62.60.0/30 is directly connected, GigabitEthernet0/0
L    45.62.60.2/32 is directly connected, GigabitEthernet0/0
S    192.168.20.0/24 [1/0] via 199.56.66.2
S    199.56.66.0/24 is variably subnetted, 2 subnets, 2 masks
C    199.56.66.0/30 is directly connected, GigabitEthernet0/1
L    199.56.66.1/32 is directly connected, GigabitEthernet0/1

R3(config)#no ip route 192.168.20.0 255.255.255.0 199.56.66.2
R3(config)#ip route 192.168.20.0 255.255.255.0 199.56.66.2
R3(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    45.62.60.0/30 is directly connected, GigabitEthernet0/0
L    45.62.60.2/32 is directly connected, GigabitEthernet0/0
S    192.168.20.0/24 [1/0] via 199.56.66.2
S    199.56.66.0/24 is variably subnetted, 2 subnets, 2 masks
C    199.56.66.0/30 is directly connected, GigabitEthernet0/1
L    199.56.66.1/32 is directly connected, GigabitEthernet0/1
```

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Top

R3

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started.

R3>enable
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip route 192.168.10.0 255.255.255.0 45.62.60.1
R3(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

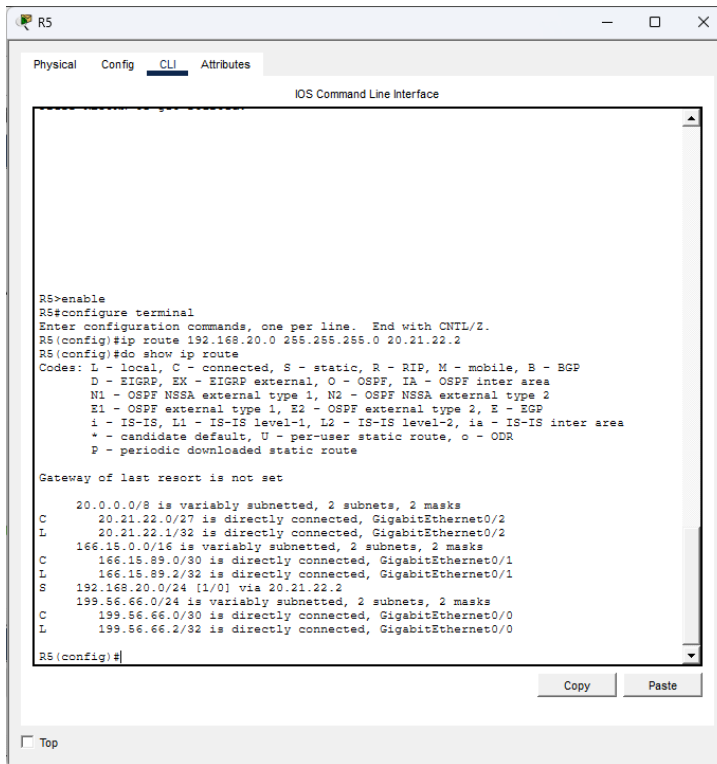
45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    45.62.60.0/30 is directly connected, GigabitEthernet0/0
L    45.62.60.2/32 is directly connected, GigabitEthernet0/0
S    192.168.10.0/24 [1/0] via 45.62.60.1
S    192.168.20.0/24 [1/0] via 199.56.66.2
S    199.56.66.0/24 is variably subnetted, 2 subnets, 2 masks
C    199.56.66.0/30 is directly connected, GigabitEthernet0/1
L    199.56.66.1/32 is directly connected, GigabitEthernet0/1

R3(config)#
```

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Top

Static routing for R5

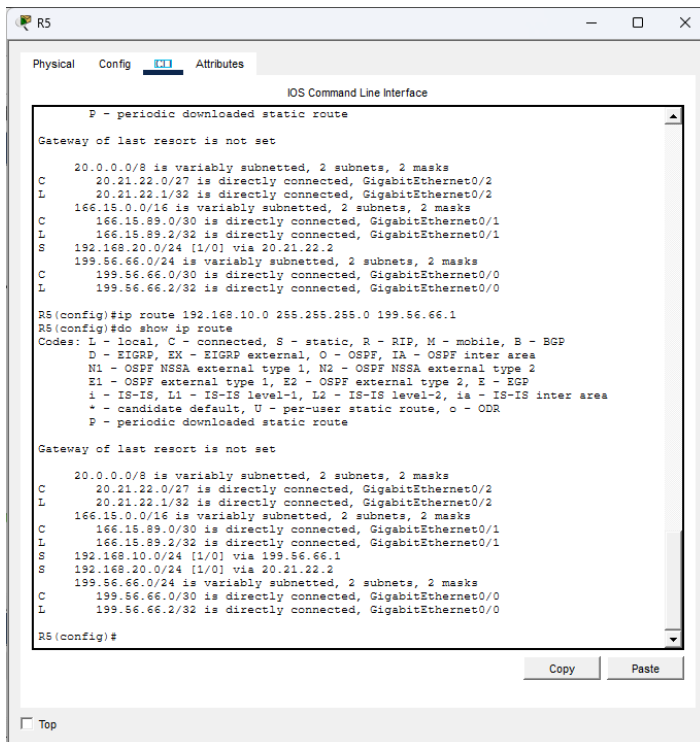


The screenshot shows the R5 CLI interface with the following commands and output:

```
R5>enable
R5#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#ip route 192.168.20.0 255.255.255.0 20.21.22.2
R5(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
    C   20.21.22.0/27 is directly connected, GigabitEthernet0/2
    L   20.21.22.1/32 is directly connected, GigabitEthernet0/2
  166.15.0.0/16 is variably subnetted, 2 subnets, 2 masks
    C   166.15.89.0/30 is directly connected, GigabitEthernet0/1
    L   166.15.89.2/32 is directly connected, GigabitEthernet0/1
    S   192.168.20.0/24 [1/0] via 20.21.22.2
  199.56.66.0/24 is variably subnetted, 2 subnets, 2 masks
    C   199.56.66.0/30 is directly connected, GigabitEthernet0/0
    L   199.56.66.2/32 is directly connected, GigabitEthernet0/0
R5(config)#
```



The screenshot shows the R5 CLI interface with the following commands and output:

```
P - periodic downloaded static route

Gateway of last resort is not set

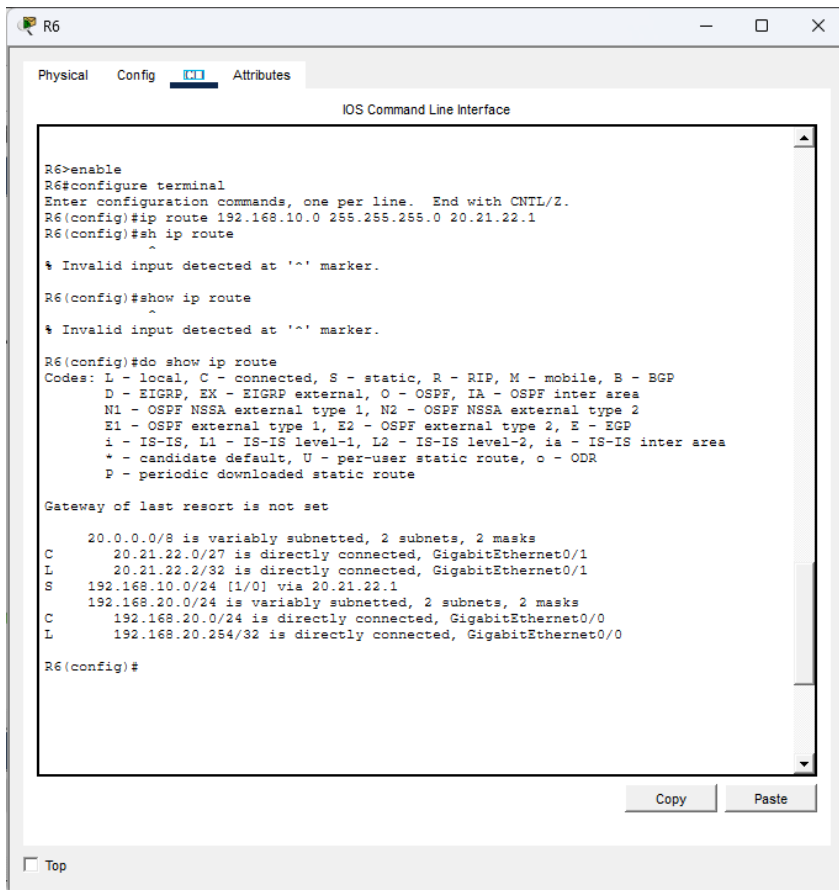
  20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
    C   20.21.22.0/27 is directly connected, GigabitEthernet0/2
    L   20.21.22.1/32 is directly connected, GigabitEthernet0/2
  166.15.0.0/16 is variably subnetted, 2 subnets, 2 masks
    C   166.15.89.0/30 is directly connected, GigabitEthernet0/1
    L   166.15.89.2/32 is directly connected, GigabitEthernet0/1
    S   192.168.20.0/24 [1/0] via 20.21.22.2
  199.56.66.0/24 is variably subnetted, 2 subnets, 2 masks
    C   199.56.66.0/30 is directly connected, GigabitEthernet0/0
    L   199.56.66.2/32 is directly connected, GigabitEthernet0/0

R5(config)#ip route 192.168.10.0 255.255.255.0 199.56.66.1
R5(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
    C   20.21.22.0/27 is directly connected, GigabitEthernet0/2
    L   20.21.22.1/32 is directly connected, GigabitEthernet0/2
  166.15.0.0/16 is variably subnetted, 2 subnets, 2 masks
    C   166.15.89.0/30 is directly connected, GigabitEthernet0/1
    L   166.15.89.2/32 is directly connected, GigabitEthernet0/1
    S   192.168.10.0/24 [1/0] via 199.56.66.1
    S   192.168.20.0/24 [1/0] via 20.21.22.2
  199.56.66.0/24 is variably subnetted, 2 subnets, 2 masks
    C   199.56.66.0/30 is directly connected, GigabitEthernet0/0
    L   199.56.66.2/32 is directly connected, GigabitEthernet0/0
R5(config)#
```

Static routing for R6



Ping successful from pc1 to pc5

